

RoboReviews

The new AI-powered product review aggregator

Presenter: Philippe

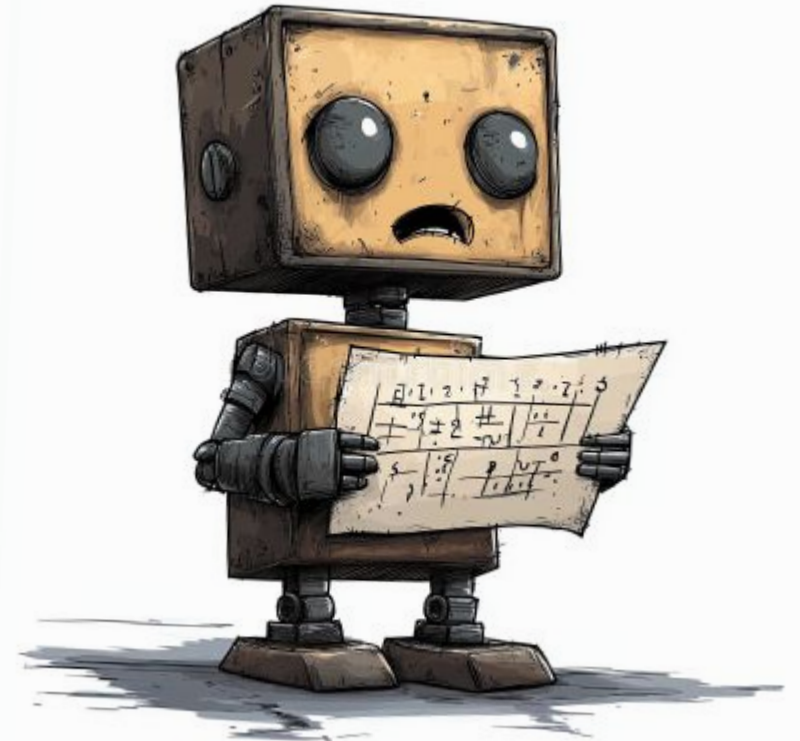


The pitch

Are you tired of reading 10,000 conflicting Amazon reviews just to buy a simple cable?

RoboReviews does the heavy lifting:

-  Ingest thousands of reviews
-  Group products
-  Analyze feelings
-  Use AI to output the ultimate "Top 3" and the "Blacklisted" products in seconds



Introduction



How they fit:

Classification scores products

Clustering groups them

Generative AI uses rankings to write the final review



Methodology:

Extracting signal from noise by combining classic NLP with modern LLMs

Methods & Data Pipeline



Dataset

Merged 3 Kaggle Amazon Consumer Reviews datasets



Preprocessing

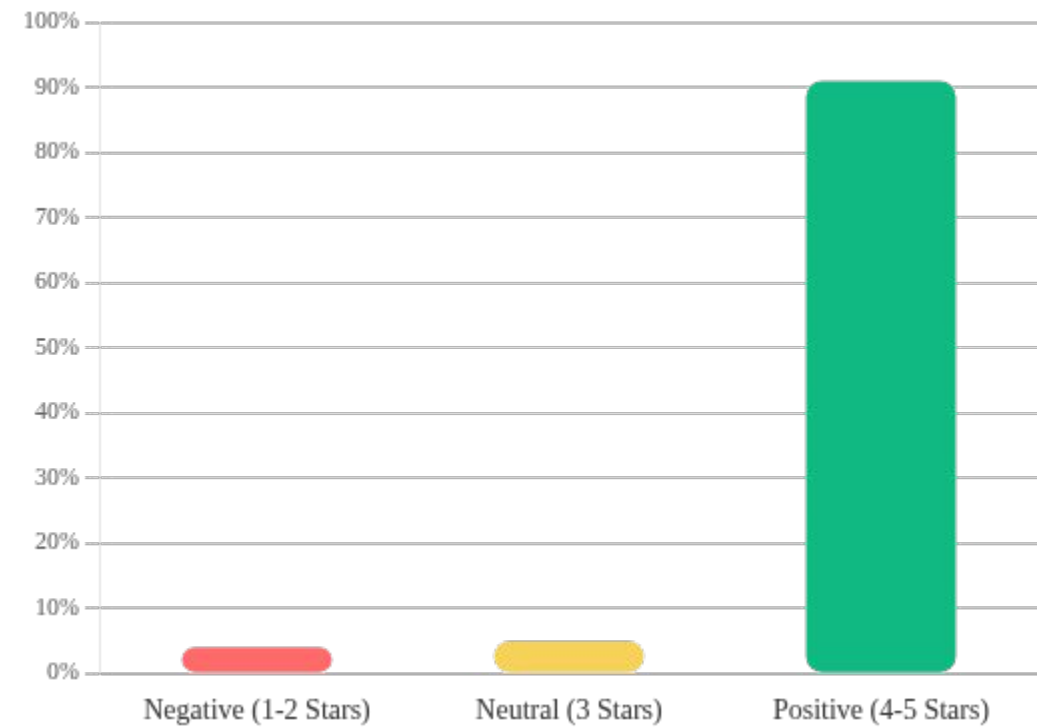
- Dropped duplicates
- Cleaned missing values
- Mapped 1–5 star ratings to a 3-class target (1–2=Negative, 3=Neutral, 4–5=Positive)



Challenge

Severe class imbalance (~91% positive reviews)

Class Imbalance Distribution



Model 1: Sentiment Classification



Goal

Classify customer reviews to give every product a mathematical quality score



What was tried

Started with `distilbert-base-uncased` but it completely failed to detect Neutral reviews (F1-score of 0)



Final Choice

Upgraded to `cardiffnlp/twitter-roberta-base-sentiment` (3 classes; fits our review format)

Model 1: Evaluation



Global Accuracy is complicated in our case (88%), so we focused on the Macro F1-Score (0.57).

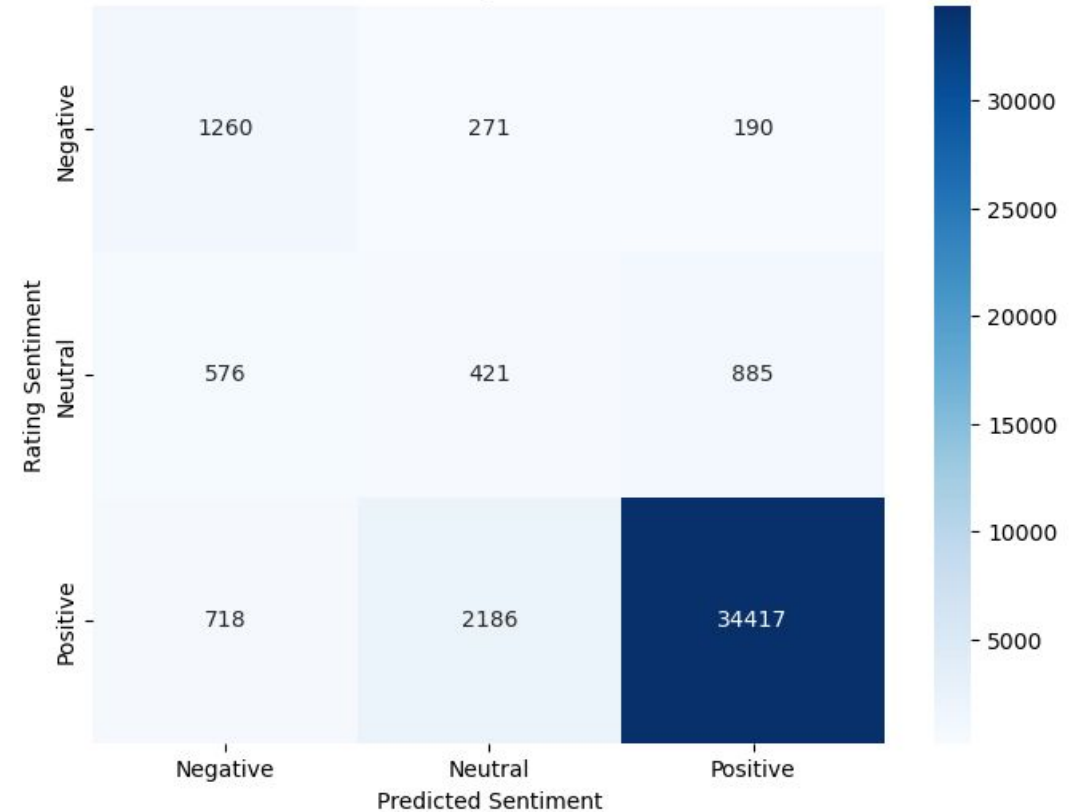


Precision on Negative reviews jumped from 20% to 49%, reducing false alarms.



Visual inspection: Many "errors" are actually human mistakes (e.g., users giving 5 stars but writing a terrible review).

Confusion Matrix: Rating vs Predicted Sentiment



Model 2: Product Categorization



Goal

Cluster products into 5 "Meta-categories" (eReaders, Batteries, Accessories, etc.)



Approach

TF-IDF vectorization + K-Means (5 clusters)



Refinement

Clustering on review text was too noisy

Switched to clustering on Product `Name` + `Categories` using n-grams (1, 3)



"Cooking the data"

Removed ubiquitous words like "Amazon"; hardcoded rules for tricky items (e.g., forcing cables into "Accessories")

Clustering Pipeline

TF-IDF

Vectorization



K-Means

Algorithm



5 Clusters

Meta-categories

Model 2: Evaluation



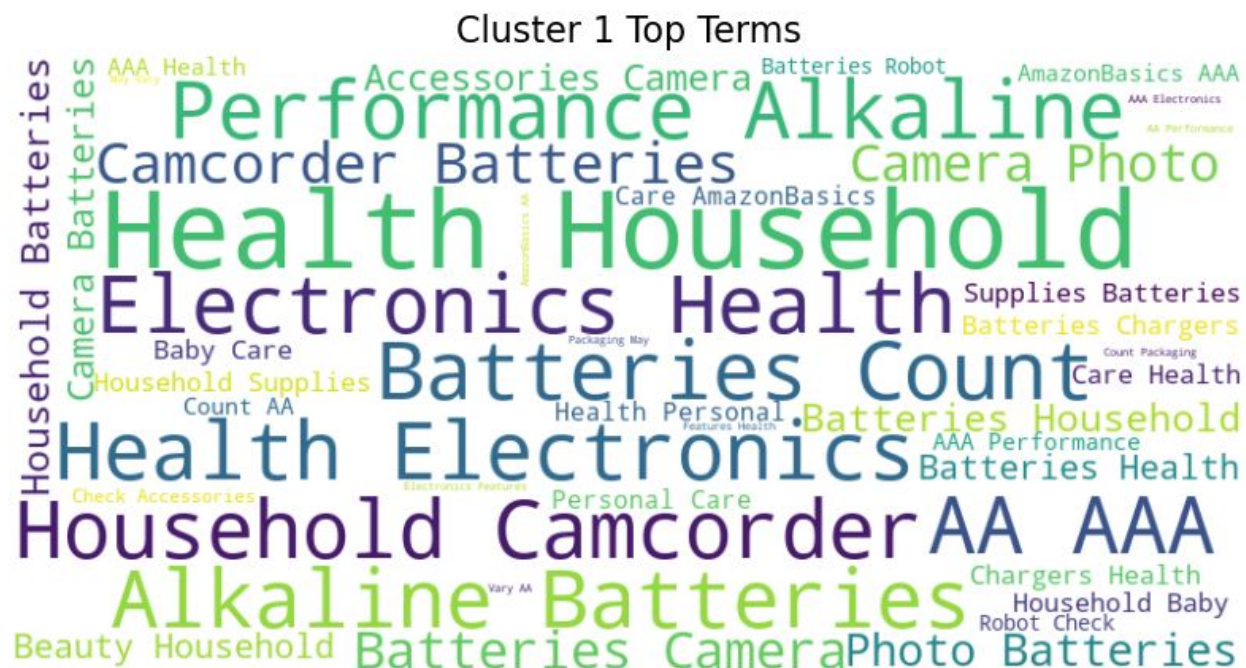
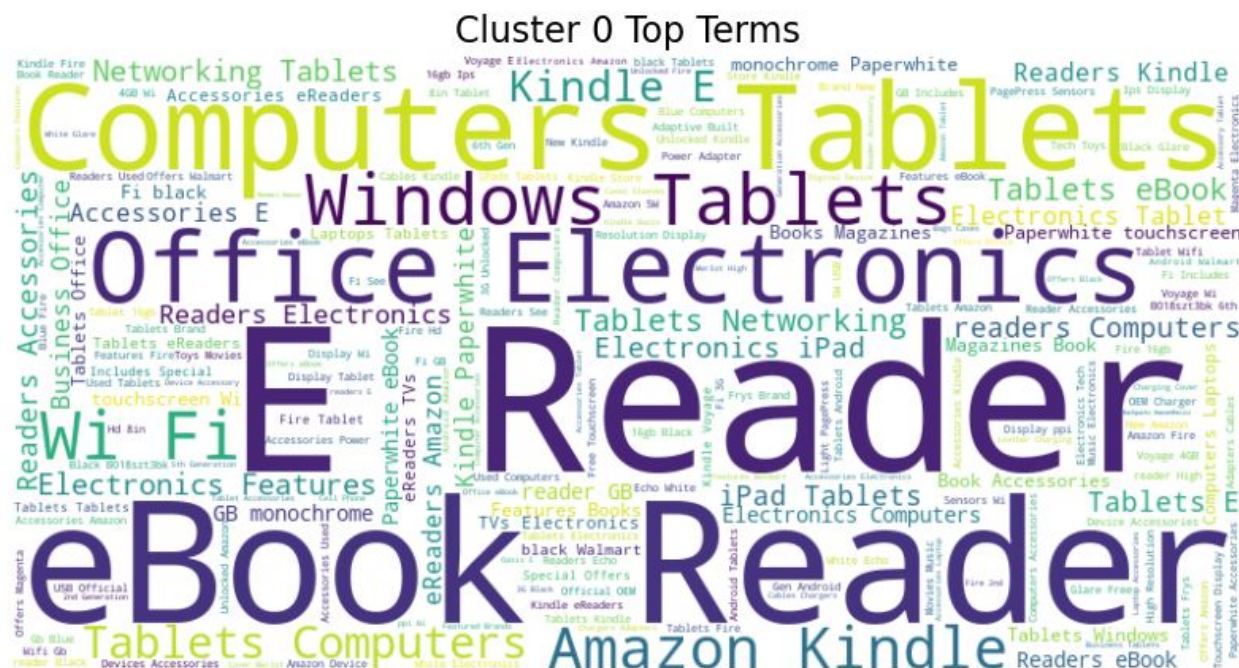
Validated using Silhouette scores and visual centroid analysis.



Hand-coded overrides heavily improved cluster coherence for edge cases.



Result: 5 highly distinct clusters ready for the recommendation engine.



Model 3: Review Summarization



Goal

Generate an article recommending the top products for each category



Model

OpenAI gpt-4o-mini via API for high readability



Prompt Engineering

Strict multi-paragraph structure to prevent hallucinations

Creativity restricted to the final concluding paragraph



Intro



Top 3 Comparison



Complaints



Blacklisted Worst Product

Model 3: Evaluation



Manual Evaluation

Readability is good; the strict prompt eliminated product hallucination.



Context injection

Passing exactly 5 real complaints into the prompt ensured factual grounding.

AI SUMMARY ARTICLE

The Ultimate Guide to Amazon Tablets: Top Picks and Pitfalls

Navigating the jungle of ads and software limitations.

RoboReviews AI 3 min read

When it comes to budget tablets, Amazon's lineup offers some enticing options. The **All-New Fire HD 8 Tablet** with its 8-inch HD Display and 16 GB of storage presents an excellent choice for casual users looking for a solid everyday device. If you're after a splash of color, the **Fire HD 8 Tablet with Alexa** in Tangerine not only provides the same specs but also integrates Alexa for hands-free assistance, making it a great pick for tech-savvy families. For those who prefer a more compact device, the **Fire Tablet with 7 Display** is a budget-friendly alternative that still delivers decent performance but may lack the visual engagement of its bigger siblings.

However, buyer beware: these devices aren't without their drawbacks. Common complaints from users revolve around the intrusive advertising and the pre-installed software that limits customization. Many users express frustration over the inability to remove ads, deeming it unacceptable for a device that is supposed to enhance user experience. Additionally, while the tablets are good for reading, the video quality has been criticized, making them less ideal for binge-watching your favorite shows. New users, especially children, have reported a steep learning curve adapting to the interface, which can detract from the overall experience.

Now, let's talk about the worst product in this category: the **Amazon - Kindle Voyage - 4GB - Wi-Fi + 3G - Black**. With its higher price point compared to the Fire tablets and a host of complaints similar to those associated with the Fire lineup, this device is a hard pass. Users have pointed out that the Kindle Voyage falls short in both functionality and flexibility, leaving many feeling like they wasted their money on a glorified e-reader that can't compete with the versatility of the Fire tablets. If you want to avoid buyer's remorse, steer clear of this one!

In conclusion, while Amazon's Fire tablets offer a mix of affordability and functionality, they're not without their quirks. It's a jungle out there in the tablet world, and as you navigate through the foliage of ads and software limitations, just remember: sometimes it's better to splurge on something that won't leave you feeling like you're stuck in a tech trap. Happy tablet hunting, and may your device not be as temperamental as your uncle during family game night!

Takeaways



Recap:

Built an end-to-end data product turning chaotic text data into structured, actionable buying advice

Live Demonstration



Let's see RoboReviews in action!



We will explore the categories, sentiment breakdowns, and read the final AI-generated verdicts

